BI:TASK-01:Exploratory Data Analysis (EDA) using Zara Sales Performance Analysis Dataset

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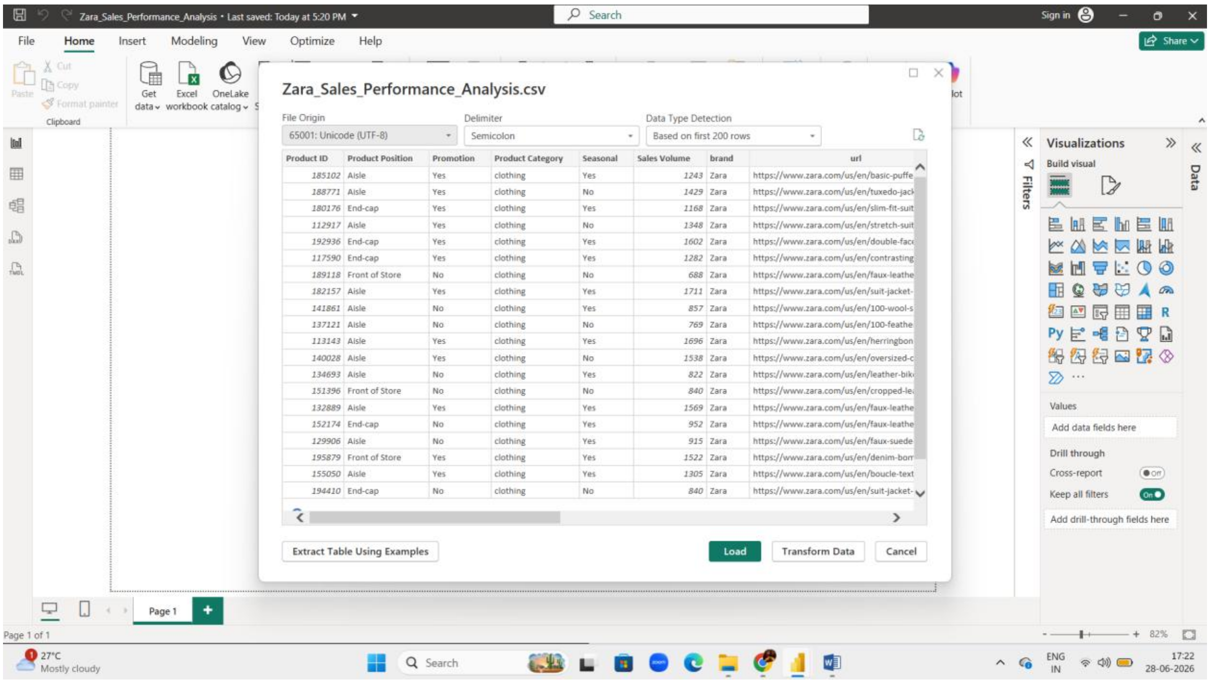
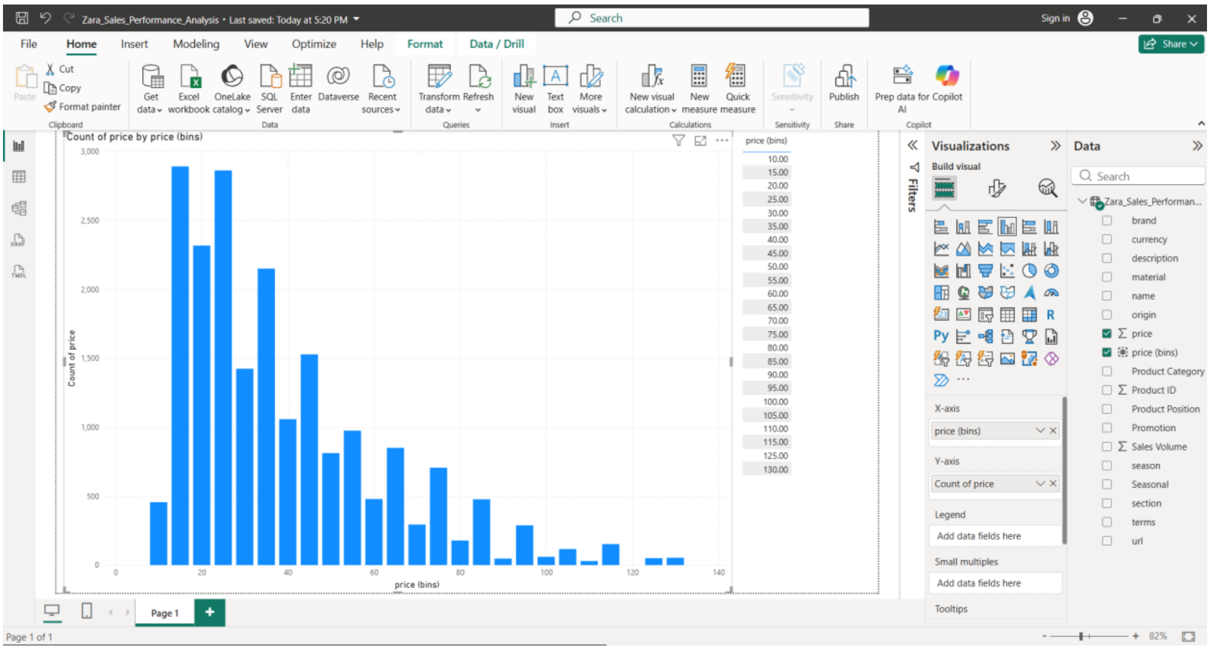
First we have to load the dataset which contain the sales data of zara

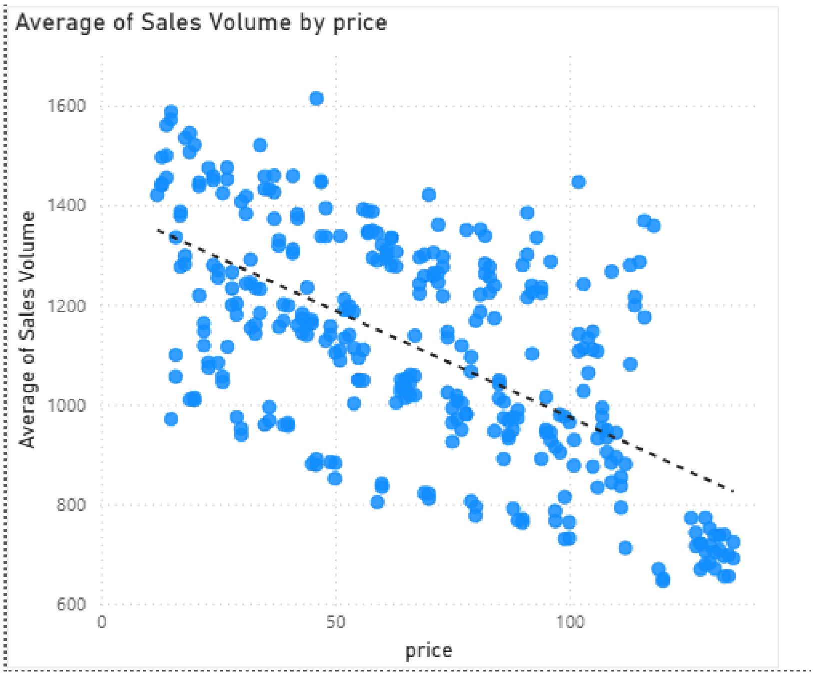


(a) What is the price range that concentrates most of Zara's products?



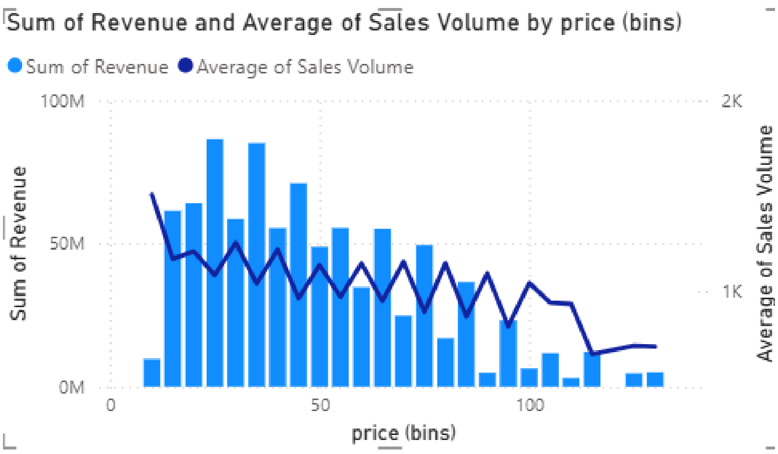
The histogram shows that Zara's products are predominantly concentrated in the lower price ranges. The highest number of products falls between ₹15–₹20 and ₹25–₹30, each containing nearly 2891 and 2862 products respectively.

(b) Do higher-priced products sell proportionally fewer units?



The scatter plot shows that products with lower prices usually have higher sales volumes, while products with higher prices generally have lower sales volumes. This indicates that higher-priced products tend to sell fewer units.

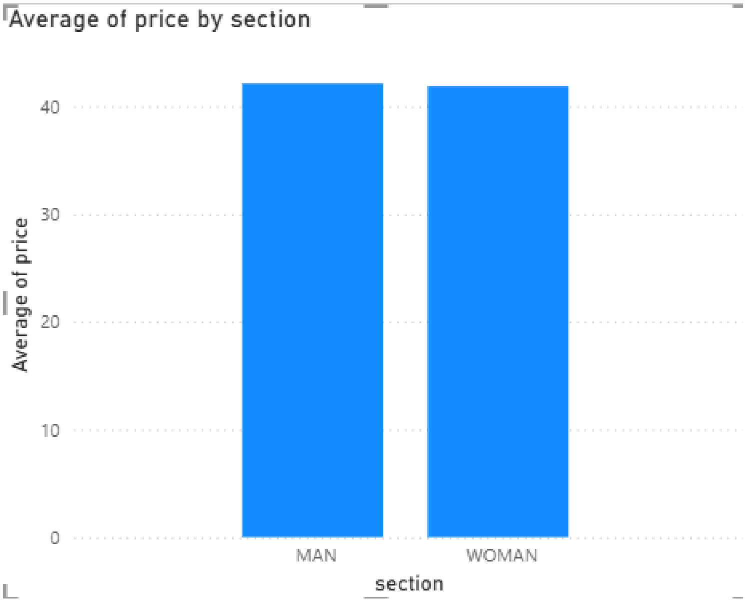
(c) Is there an "ideal" price range that balances sales volume and total revenue?



The chart suggests that the ₹20–₹35 price range provides the best balance between sales volume and total revenue. Products in this range generate the highest revenue while maintaining relatively high average sales volume. Lower-priced products achieve good sales volume but contribute less revenue per product, whereas higher-priced products experience a decline in both revenue and average sales volume.

(d) Does the average price differ significantly between sections (MAN, WOMAN)?

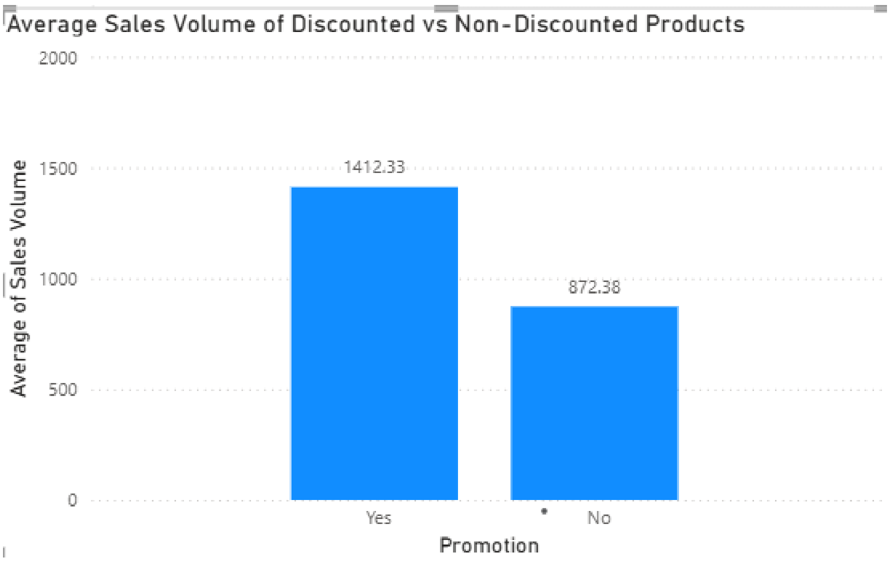
The average prices of products in the MAN and WOMAN sections are nearly identical. Although the MAN(42.13) section has a slightly higher average price than women(41.86), the difference is very small. Therefore, there is no significant difference in the average product prices between the two sections.



(e) Do discounted products significantly reduce their average price?

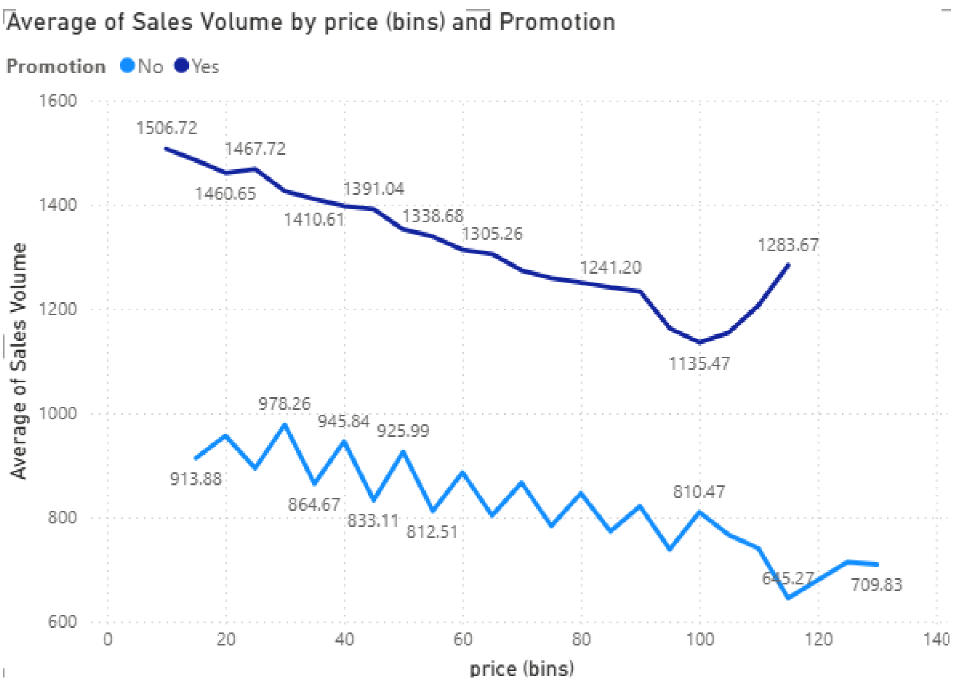
Discounted products have a lower average price than non-discounted products. The average price decreases from 45 to 38, indicating that promotions effectively reduce the average selling price. Therefore, discounted products are generally sold at lower prices than non-discounted products.

(f) Do discounted products have, on average, higher sales volumes?



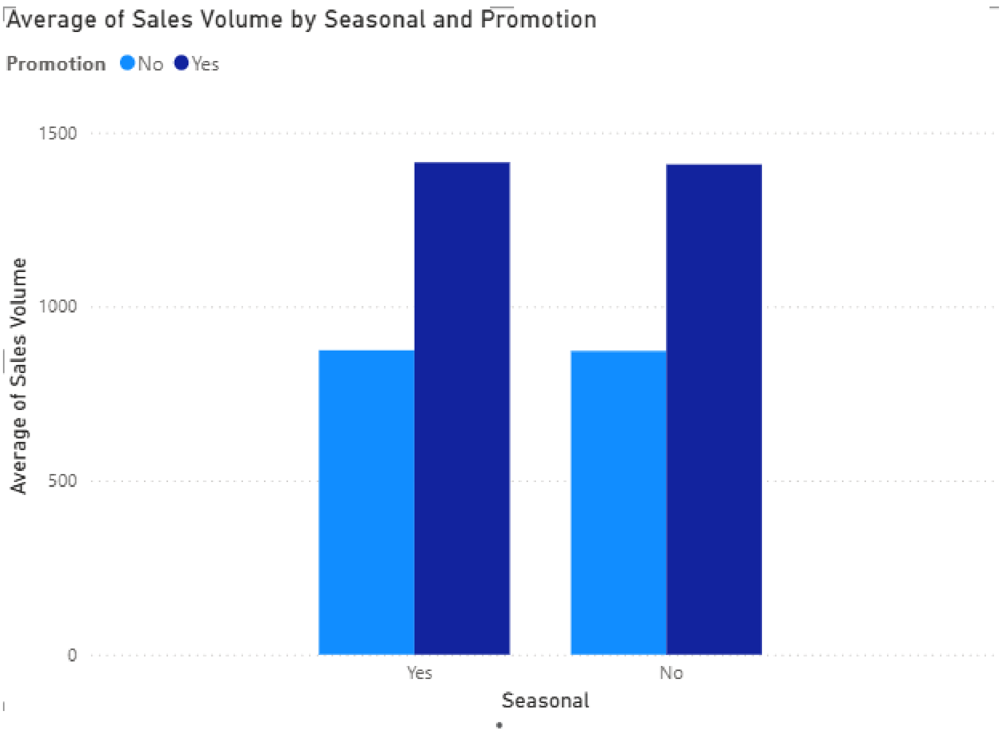
The chart indicates that discounted products achieve a significantly higher average sales volume than non-discounted products. Products on promotion sell an average of 1412.33 units, while products without promotion sell 872.38 units on average. This suggests that discounts have a positive impact on customer demand and increase sales volume.

(g) Is the difference in sales volume between discounted and non-discounted products greater within a specific price range?



The line chart shows that discounted products consistently have higher average sales volumes than non-discounted products across almost all price ranges. The difference is greatest in the lower and mid-price ranges (approximately ₹15–₹50), where promotional products sell significantly more units. As the price increases, the gap becomes slightly smaller, but discounted products continue to outperform non-discounted products in terms of average sales volume. This suggests that promotions are most effective for products in the lower and mid-price ranges.

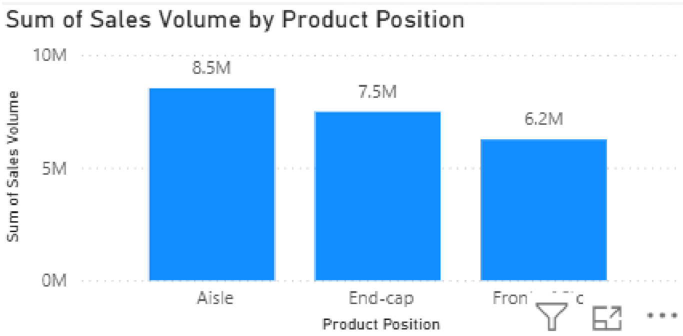
(h) Do promotions affect seasonal products more than non-seasonal ones?

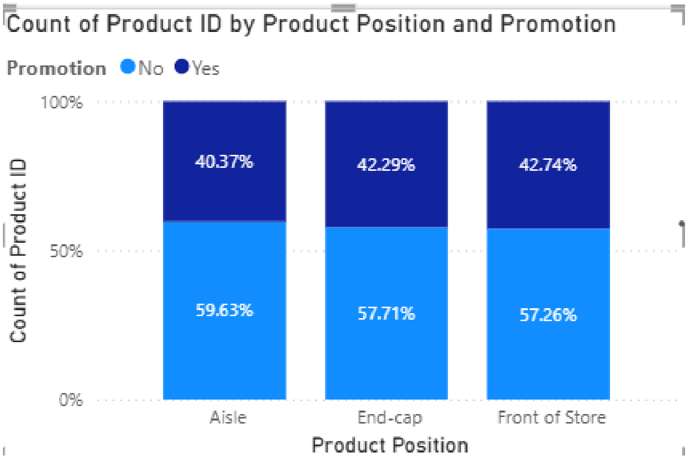


Promotions increase the average sales volume for both seasonal and non-seasonal products. However, the difference between promoted and non-promoted products is almost the same in both groups. Therefore, promotions do **not** appear to have a stronger effect on seasonal products than on non-seasonal products. They improve sales volume by a similar amount regardless of seasonality.

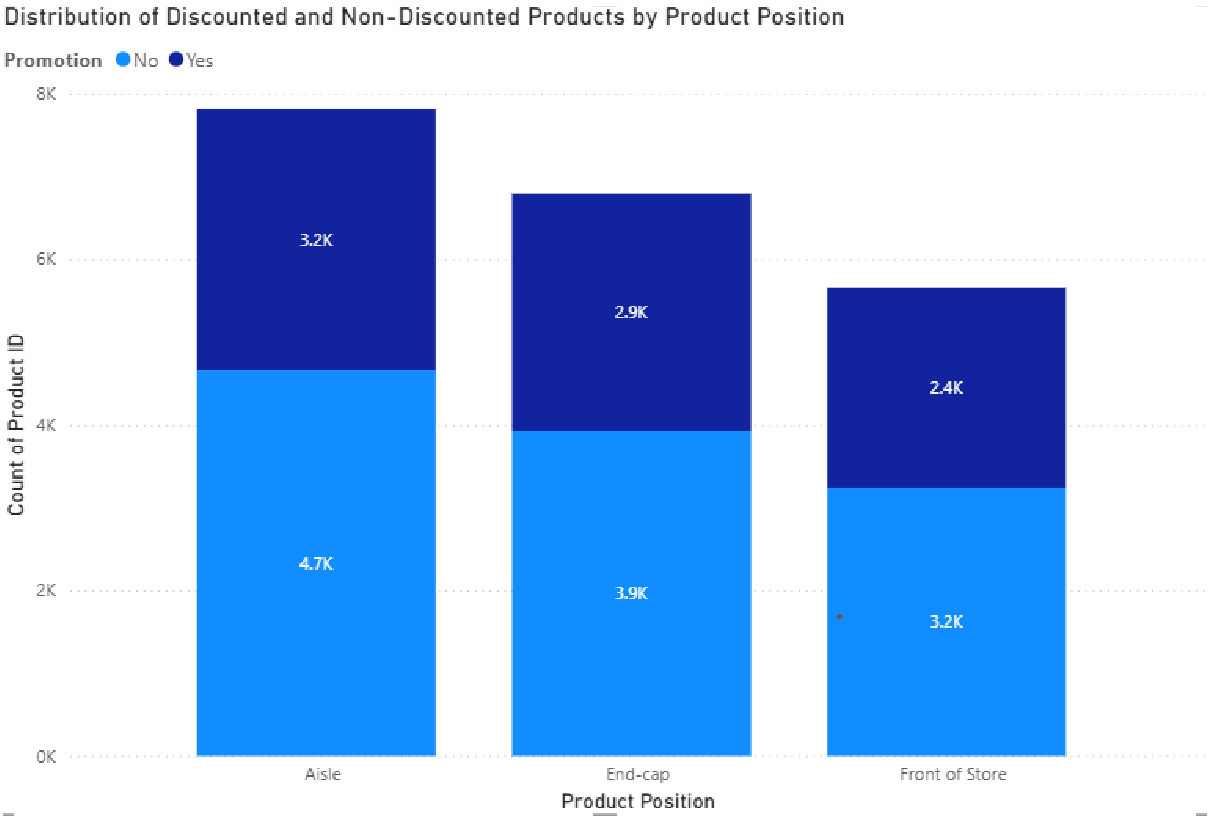
(i) Is there any relationship between the percentage of discounted products and total sales?

The comparison between the percentage of discounted products and total sales does not show a strong positive relationship. Although the Front of Store has the highest percentage of discounted products (42.74%), it records the lowest total sales volume (6.2M). In contrast, the Aisle has the lowest percentage of discounted products (40.37%) but achieves the highest total sales (8.5M). Therefore, the results suggest that a higher percentage of discounted products does not necessarily lead to higher total sales, and other factors such as product placement or customer traffic may also influence sales performance.





(j) Do discounted products appear more frequently in a specific Product Position (e.g., End-cap or Front of Store)?



Discounted products are found in all product positions, but they appear most frequently in the Aisle, followed by the End-cap, and least frequently at the Front of Store. This suggests that the Aisle is the most common location for discounted products. However, non-discounted products are also more numerous in every product position, indicating that promotions are distributed across all locations rather than being concentrated in a single display

Final Dashboard